

Mathematical Modeling of Influenza Using SIR and SIRS Models

Final Project Report

Course: Mathematical Modeling

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1. Introduction

Influenza is a highly contagious viral disease that spreads rapidly through human populations. Because immunity to influenza often wanes over time, individuals may become susceptible again after recovering. This makes influenza a good example of a disease that can be modeled using both the SIR and SIRS frameworks. In this project, I investigated how key epidemiological parameters affect disease dynamics: infection rate (β), recovery rate (γ), immunity loss rate (ω), and vaccination rate (v). Using computational simulations, I analyzed how outbreaks change under different conditions and compared permanent immunity to temporary immunity.

2. Methods

Two mathematical models were used: SIR Model: $dS/dt = -\beta SI$ $dI/dt = \beta SI - \gamma I$ $dR/dt = \gamma I$ SIRS Model: $dS/dt = -\beta SI + \omega R$ $dI/dt = \beta SI - \gamma I$ $dR/dt = \gamma I - \omega R$ Vaccination was included as a transfer from S to R at rate v . Initial conditions were: $S(0) = 9990$, $I(0) = 10$, $R(0) = 0$ with total population $N = 10,000$. The simulations were run for 200 days using numerical solvers in R.

3. Results

The simulation results show that higher infection rates lead to faster and larger outbreaks. In the SIR model, infections eventually decline to zero because immunity is permanent. In the SIRS model, immunity loss causes the disease to persist at a stable endemic level. Vaccination reduces both the peak number of infected individuals and the long-term equilibrium level. Higher immunity loss increases long-term disease persistence.

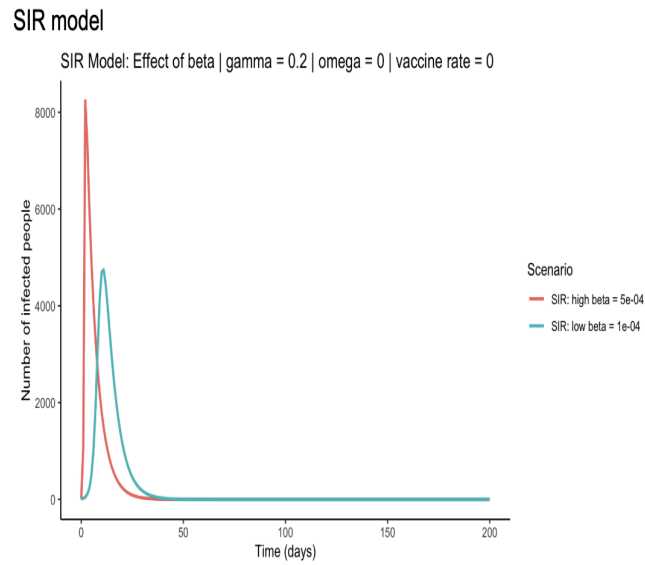


Figure: Simulation output from SIR/SIRS models.

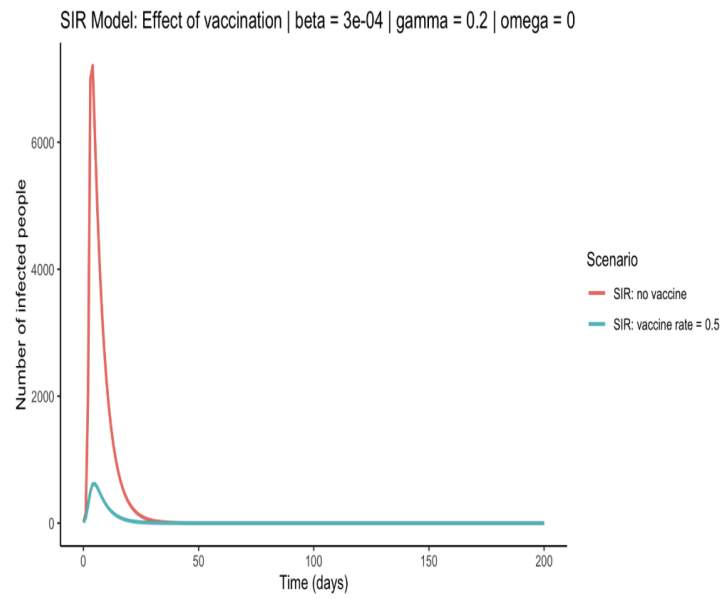


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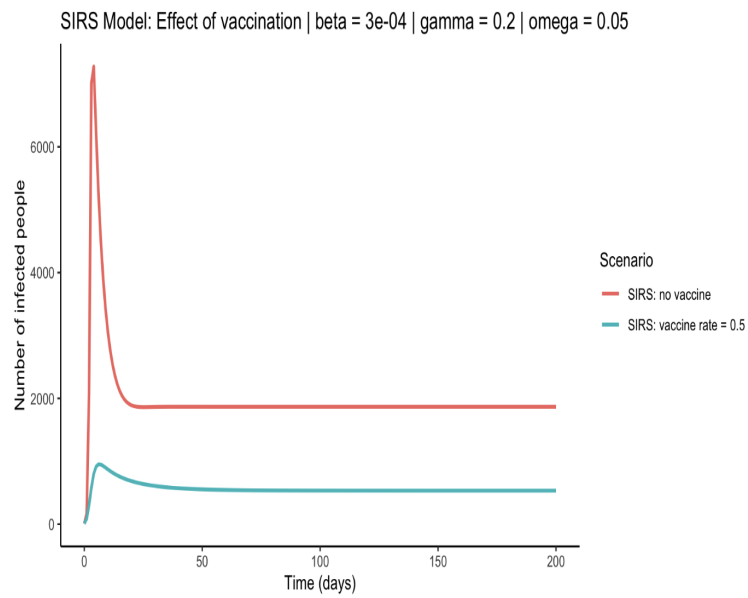


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4. Discussion

The results demonstrate that the infection rate β is the strongest driver of outbreak severity. Small increases in β caused significantly higher peaks in infections. Temporary immunity in the SIRS model explains why influenza continues to circulate yearly. Vaccination proved to be the most effective way to control outbreaks.

5. Conclusion

This project shows how mathematical models can be used to understand influenza dynamics. SIR and SIRS models provide important insights into transmission, immunity, and vaccination. The findings strongly support vaccination as the most effective public health intervention.

6. Bibliography

Kermack, W., & McKendrick, A. (1927). A Contribution to the Mathematical Theory of Epidemics.
Hethcote, H. (2000). The Mathematics of Infectious Diseases. SIAM Review. CDC (2024). Influenza
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